

Technical provenance

What stands behind the software: data lineage, leakage controls, model design, calibration, evaluation, reporting policy, and reproducible artifacts.

RESEARCH PROTOTYPE. Decision support only. Every result must be confirmed with standard laboratory susceptibility testing.

E. COLI GENOMES

3,000

MASH GROUPS

1,548

AMR FEATURES

460

DRUG MODELS

4

System at a glance

- Input: one reconstructed, quality-checked *Escherichia coli* FASTA.
- Features: AMRFinderPlus gene and point-mutation findings mapped to a fixed, versioned binary schema.
- Prediction: one regularized scikit-learn logistic-regression model per antibiotic.
- Output: likely to fail, likely to work, or no-call; calibrated confidence; evidence category; mandatory lab-confirmation flag.
- Frontend: Streamlit clinical-style antibiogram consuming one stable backend adapter.

Supported panel

Ciprofloxacin, Ceftriaxone, Gentamicin, Ampicillin

Deliberate scope boundaries

No sampling, sequencing, assembly, species identification, treatment automation, agent, or LLM. The model is species-specific and is not a clinical diagnostic device.

Data and leakage control

Training source

BV-BRC bacterial genomes paired with laboratory resistance phenotypes. The frozen cohort contains 3,000 quality-filtered *E. coli* genomes. Missing drug labels are excluded per drug; they are never imputed. Labels use the explicit Resistant/Susceptible vocabulary. Intermediate is excluded rather than silently merged.

Homology-aware grouping

Mash MinHash sketches use $k=21$, sketch size 10,000, and distance threshold 0.001. Genomes connected at or below the threshold form one group. The threshold is configurable. Group IDs, qualifying edges, and the leakage audit are retained as evidence.

Partition	Genomes	Purpose
Train	2,100	Fit and tune regularized logistic models using group-aware CV
Calibration	452	Fit Platt/sigmoid calibrators and select abstention thresholds
Hidden test	448	Final untouched evaluation; never used for fitting or threshold selection

Leakage audit: PASSED - 1,548 genetic groups; no group spans partitions.

Why this matters

Random genome splits can place near-identical isolates in training and testing, inflating performance. Genome Firewall makes the group boundary a hard invariant and fails training if homology groups cross splits.

Feature contract

Schema ID: gf-amr-v1-819928b0365edbd2. The matrix is sparse and binary. Precomputed AMRFinder TSV and ready-made matrices are supported; matrices must provide a matching schema ID and exact ordered columns, or validation fails loudly.

Model, calibration, and calling policy

Stage	Implementation	Guardrail
Model	One L2-regularized logistic regression per drug; CPU-only	Group-aware tuning on training groups
Calibration	Held-out sigmoid/Platt map per drug	Calibration genomes are not used to fit the base model
Thresholds	Drug-specific resistant and susceptible cutoffs	Middle band produces no-call
Evidence	Known marker / statistical association / no known signal	Known-marker claims require curated mapping
Intrinsic gate	Optional species/drug configuration	No target absence inferred from AMRFinder

Conservative susceptibility policy

Absence of an AMRFinder resistance marker is not treated as proof of susceptibility. Under the default policy, marker-free low-risk predictions prefer no-call rather than a confident likely-to-work statement. A deterministic forced likely-to-fail result is allowed only for explicitly configured, scientifically defensible intrinsic-resistance rules.

Current evidence limitation

The shipped bundle does not include a reviewed marker_drug_map.json. Therefore the software conservatively reports model-supported results as statistical evidence and does not overstate causal gene evidence. Adding a curated mapping is a documented future improvement, not a hidden assumption.

Reproducibility

The bundle freezes the feature schema, drug panel, label policy, per-drug models, calibrators, thresholds, training metadata, and leakage audit. Random state is 42. AMRFinderPlus 4.2.7 used database 2026-05-15.1. Pipeline commands are resumable and Slurm wrappers are provided; model training itself does not need a GPU.

Hidden-test evidence

All values below come from the untouched homology-grouped hidden test. Metrics are computed per drug after excluding genomes with missing labels for that drug.

Drug	N	Bal. acc.	R recall	S recall	AUROC	PR-AUC	Brier	No-call	Called acc.
Ciprofloxacin	418	0.977	0.957	0.996	0.993	0.992	0.020	0.081	0.990
Ceftriaxone	144	0.848	0.922	0.775	0.965	0.972	0.083	0.174	0.975
Gentamicin	417	0.944	0.909	0.980	0.959	0.895	0.038	0.010	0.964
Ampicillin	340	0.892	0.946	0.839	0.965	0.992	0.053	0.226	0.992

Metric definitions

- Balanced accuracy: mean of resistant recall and susceptible recall.
- AUROC and PR-AUC: threshold-free discrimination using calibrated resistance probability.
- Brier score and reliability curves: probability calibration quality; lower Brier is better.

- No-call rate: fraction inside the abstention band. Accuracy-on-called excludes no-calls and must be read together with coverage.

Validation performed

62 automated backend, adapter, grouping, schema, training-roundtrip, and presentation tests passed. A real held-out E. coli FASTA was processed through AMRFinder, the packaged model bundle, report translation, and the Streamlit adapter. The real-mode Streamlit health endpoint returned OK.

Traceability and artifacts

Artifact	Repository location
Packaged model bundle	genome-firewall/models/ecoli-bundle/
Detailed hidden-test metrics	genome-firewall/models/ecoli-bundle/metrics/summary.json
Leakage audit	genome-firewall/models/ecoli-bundle/leakage_audit.json
Pipeline and cluster runbook	genome-firewall/cluster/README.md
End-to-end handoff	HANDOFF_CHECKLIST.md
Merged release commit	3947ea5 - main

Interpretation

These results support a hackathon research prototype for E. coli and the configured four-drug panel. They do not establish prospective clinical validity, transportability to other populations, or suitability for treatment decisions without laboratory confirmation and professional review.

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